

Dhanush M

Data Scientist · ML Engineer — LLM & Applied AI

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Takes production computer-vision and LLM systems from zero to deployment — two now live on client sites: automated safety and quality inspection (~86% precision, entirely on CPU) and a multilingual voice-to-structured-data service (form entry cut from minutes to under one). Expertise in object detection and inference optimisation under hard cost constraints, multi-agent LLM and vision-language pipelines, and the multi-tenant data platforms serving them.

EXPERIENCE

Slate Technologies — Construction Technology Feb 2025 – Present

Assistant Software Engineer

Visual Intelligence — Automated Site Safety & Quality Inspection BUILT FROM SCRATCH

- Delivered drone-footage hazard and defect detection to production sites with no GPU budget — inference had to run on CPU-only EC2, so I iterated the model across three architectures (SAHI-tiled YOLO → RF-DETR → TorchVision Faster R-CNN) and tuned tiled inference to reach ~86% precision (mAP 0.65–0.70) on 4K imagery inside that constraint.
- Made CPU inference viable by cutting the work ~99%: frame selection, ORB feature-similarity deduplication and a resolution-independent blur gate reduce a one-minute 4K clip from ~1,800 frames to the 6–7 genuinely unique ones worth detecting on; consolidated safety and quality into a single pass served via sync and batched async APIs.
- Turned raw boxes into findings a site manager can act on: a vision-language pass maps each detection onto a deterministic CSI-code taxonomy, and a six-stage LLM agent chain converts each project's own uploaded safety documents into detection rules — so flags reflect that site's requirements instead of a fixed generic rule set.

STACK Python · PyTorch / TorchVision · RF-DETR · YOLO + SAHI · OpenCV · FastAPI · OpenAI API · PostgreSQL + pgvector · AWS S3, Lambda

Voice-to-Form-Fill — Multilingual LLM Speech-to-Structured-Data SOLE AUTHOR

- Replaced on-site form typing with dictation, cutting entry from 2–5 minutes to under one — a two-agent LLM pipeline matches spoken intent to the tenant's live form types, then extracts field values against that form's real schema, fronted by a normalization stage that accepts code-mixed Tamil/Hindi/English so engineers speak naturally rather than in formal English.
- Halved response time from 10–12s to 5–6s by running the form-selection LLM call and the database schema prefetch concurrently on a thread pool — on site, every extra second of waiting pushes users back to typing. Deterministic post-processing resolves spoken names to internal entity IDs and discards unmatched values, so a mis-heard name never writes bad data into the platform.

STACK Python · Flask · OpenAI API (sync + async) · asyncio · ThreadPoolExecutor · PostgreSQL

Document Intelligence — Drawing & Specification Extraction

- Made client document sets searchable instead of manually indexed — each client onboards thousands of drawing and specification PDFs. Hierarchical LLM agents classify sheets by discipline and extract set/version data; table-of-contents-aware parsing splits spec books into sections.
- Authored the shared ingestion layer (relational, graph, and vector clients) that both parsers write into, feeding one retrieval index.

Analytics Data Platform & Multi-Tenant APIs

- Closed cross-tenant exposure risk on an analytics API serving several client organisations — JWT-verified tenant/project scoping that rejects mismatched parameters, plus AES-256 encryption of stored queries; designed the underlying multi-tenant schema (data views, dashboards, chart definitions with role-based permissions) and its reporting views.
- Removed polling from media ingestion with in-database triggers that queue site uploads and fire webhooks straight into the CV service — and returned retry/timeout (15s→60s, 60s→240s) once CPU-bound inference began outrunning the original limits and failing jobs mid-run.

AI Site Planning

- Fixed layout generation that was placing buildings overrunning their plots — replaced a fixed max-height heuristic with dimension-based 3D model matching (model-dimension lookup plus a reworked classification prompt) so footprints resolve to models that physically fit; also shipped an automated site flythrough with spline-interpolated camera pathing.

SKILLS

Languages	Python · SQL
ML & CV	PyTorch · TorchVision · RF-DETR · YOLO / SAHI tiled inference · OpenCV · object detection · CPU-only inference optimization
LLM & Applied AI	OpenAI API · vision-language (VLM) pipelines · multi-agent pipeline design · prompt engineering · structured extraction · taxonomy mapping
Data	PostgreSQL · pgvector · Neo4j · vector / graph / full-text retrieval · Hasura GraphQL · PostGIS · pandas
Platform	FastAPI · Flask · AWS S3 & Lambda · Docker · deployment scripting for DevOps handoff

EDUCATION

BE, Computer Science and Engineering — Sona College of Technology 2021 – 2025